

$\mathcal{M}_A \setminus \mathcal{M}_B$	$K = 1$	$K = 2$	$K = 3$	$K = 4$
$K = 1$				
$K = 2$	3.19			
$K = 3$	11.52	3.62		
$K = 4$	11.97	3.76	1.04	

Table 4:: Bayes Factors for Gaussian mixture model example.

$\mathcal{M}_A \setminus \mathcal{M}_B$	$K = 1$	$K = 2$	$K = 3$	$K = 4$
$K = 1$		1.37	1.84	2.23
$K = 2$	0.73		1.35	1.63
$K = 3$	0.54	0.74		1.21
$K = 4$	0.45	0.61	0.83	

Table 5:: Bayes Factors for the multinomial mixture model.

Appendix C: Proof of Proposition 1

The PPN compares the distributions: $p_A(d_B(\mathbf{y}_{\text{rep}}^A; \mathbf{y}_{\text{val}}) | \mathbf{y}_{\text{in}}, \mathbf{X}_{\text{in}})$ and $p_B(d_B(\mathbf{y}_{\text{rep}}^B; \mathbf{y}_{\text{val}}) | \mathbf{y}_{\text{in}}, \mathbf{X}_{\text{in}})$. The posterior predictive distribution of model A is:

$$y_{\text{rep},i}^A | \mathbf{y}_{\text{in}} \sim N(\bar{\mathbf{y}}_{\text{in}}, 2). \quad (47)$$

The posterior predictive distribution of model B is:

$$y_{\text{rep},i}^B | \mathbf{y}_{\text{in}} \sim N(\hat{\theta}_{\text{in}}^B + \mathbf{x}_{\text{in},i}^\top \hat{\beta}_{\text{in}}^B, 2 + \mathbf{x}_{\text{in},i}^\top (\mathbf{X}_{\text{in}}^\top \mathbf{X}_{\text{in}})^{-1} \mathbf{x}_{\text{in},i}), \quad (48)$$

where $\hat{\beta}_{\text{in}}^B = (\mathbf{X}_{\text{in}}^\top \mathbf{X}_{\text{in}})^{-1} \mathbf{X}_{\text{in}}^\top \mathbf{y}_{\text{in}}$ and $\hat{\theta}_{\text{in}}^B = \bar{\mathbf{y}}_{\text{in}} - \bar{\mathbf{X}}_{\text{in}} \hat{\beta}_{\text{in}}^B$.

We first consider the distribution of data from model A under the model B diagnostic:

$$d_B(\mathbf{y}_{\text{rep}}^A; \mathbf{y}_{\text{val}}) = \sum_{i=1}^n [y_{\text{rep},i}^A - \hat{\theta}_{\text{val}}^B - \mathbf{x}_{\text{in},i}^\top \hat{\beta}_{\text{val}}^B]^2 \quad (49)$$

$$= \sum_{i=1}^n [y_{\text{rep},i}^A - \bar{\mathbf{y}}_{\text{in}} - (\hat{\theta}_{\text{val}}^B - \bar{\mathbf{y}}_{\text{in}}) - \mathbf{x}_{\text{in},i}^\top (\hat{\beta}_{\text{val}}^B - \beta_0)]^2 \quad (50)$$

where $\beta_0 = \mathbf{0}_p$ is the true parameter value. As $y_{\text{rep},i}^A - \bar{\mathbf{y}}_{\text{in}} \sim N(0, 2)$, we have:

$$d_B(\mathbf{y}_{\text{rep}}^A; \mathbf{y}_{\text{val}}) = \sum_{i=1}^n [\sqrt{2} \cdot Z_i - (\hat{\theta}_{\text{val}}^B - \bar{\mathbf{y}}_{\text{in}}) - \mathbf{x}_{\text{in},i}^\top (\hat{\beta}_{\text{val}}^B - \beta_0)]^2, \quad (51)$$

where $Z_i \stackrel{\text{ind}}{\sim} N(0, 1)$. As $n \rightarrow \infty$, we have:

$$(\hat{\theta}_{\text{val}}^B - \theta_0 - (\bar{\mathbf{y}}_{\text{in}} - \theta_0)) \sim N(0, 2/n). \quad (52)$$

We have

$$\mathbf{x}_{\text{in},i}^\top (\hat{\beta}_{B,\text{val}} - \beta_0) \sim N(0, \mathbf{x}_{\text{in},i}^\top [\mathbf{X}_{\text{val}}^\top \mathbf{X}_{\text{val}}]^{-1} \mathbf{x}_{\text{in},i}). \quad (53)$$

By Eq. 20, we have $\mathbf{x}_{\text{in},i}^\top [\mathbf{X}_{\text{val}}^\top \mathbf{X}_{\text{val}}]^{-1} \mathbf{x}_{\text{in},i} \rightarrow p/n$ as $n \rightarrow \infty$.

Then, as $n \rightarrow \infty$ and $p/n \rightarrow 0$, $d_B(\mathbf{y}_A)$ has an approximately $2\chi_n^2$ -distribution (to first order).

We now consider the distribution of data from $\mathcal{M}_B$ under the $\mathcal{M}_B$ diagnostic:

$$d_B(\mathbf{y}_{\text{rep}}^B; \mathbf{y}_{\text{val}}) = \sum_{i=1}^n [y_{\text{rep},i}^B - \hat{\theta}_{\text{val}}^B - \mathbf{x}_{\text{in},i}^\top \hat{\beta}_{\text{val}}^B]^2 \quad (54)$$

$$= \sum_{i=1}^n [y_{\text{rep},i}^B - \hat{\theta}_{\text{in}}^B - \mathbf{x}_{\text{in},i}^\top \hat{\beta}_{\text{in}}^B + (\hat{\theta}_{\text{in}}^B - \hat{\theta}_{\text{val}}^B) + \mathbf{x}_{\text{in},i}^\top (\hat{\beta}_{\text{in}}^B - \hat{\beta}_{\text{val}}^B)]^2 \quad (55)$$

$$= \sum_{i=1}^n \left[\sqrt{2 + \mathbf{x}_{\text{in},i}^\top (\mathbf{X}_{\text{in}}^\top \mathbf{X}_{\text{in}})^{-1} \mathbf{x}_{\text{in},i}} \cdot \tilde{Z}_i + (\hat{\theta}_{\text{in}}^B - \hat{\theta}_{\text{val}}^B) + \mathbf{x}_{\text{in},i}^\top (\hat{\beta}_{\text{in}}^B - \hat{\beta}_{\text{val}}^B) \right]^2 \quad (56)$$

where $\tilde{Z}_i \sim N(0, 1)$. As $n \rightarrow \infty$, we have:

$$(\hat{\theta}_{\text{in}}^B - \hat{\theta}_{\text{val}}^B) \sim N(0, 2/n). \quad (57)$$

Further,

$$\mathbf{x}_{\text{in},i}^\top (\hat{\beta}_{\text{in}}^B - \hat{\beta}_{\text{val}}^B) \sim N(0, \mathbf{x}_{\text{in},i}^\top [\mathbf{X}_{\text{val}}^\top \mathbf{X}_{\text{val}}]^{-1} \mathbf{x}_{\text{in},i} + \mathbf{x}_{\text{in},i}^\top [\mathbf{X}_{\text{in}}^\top \mathbf{X}_{\text{in}}]^{-1} \mathbf{x}_{\text{in},i}). \quad (58)$$

As $n \rightarrow \infty$, we have $\mathbf{x}_{\text{in},i}^\top [\mathbf{X}_{\text{val}}^\top \mathbf{X}_{\text{val}}]^{-1} \mathbf{x}_{\text{in},i} + \mathbf{x}_{\text{in},i}^\top [\mathbf{X}_{\text{in}}^\top \mathbf{X}_{\text{in}}]^{-1} \mathbf{x}_{\text{in},i} \rightarrow 2p/n$.

Then, as $n \rightarrow \infty$ and $p/n \rightarrow 0$, we have $d_B(\mathbf{y}_{\text{rep}}^B; \mathbf{y}_{\text{val}})$ has an approximately $2\chi_n^2$ -distribution (to first order).

Thus, $d_B(\mathbf{y}_{\text{rep}}^B; \mathbf{y}_{\text{val}})$ is asymptotically equal in distribution to $d_B(\mathbf{y}_{\text{rep}}^A; \mathbf{y}_{\text{val}})$ (to first order).

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